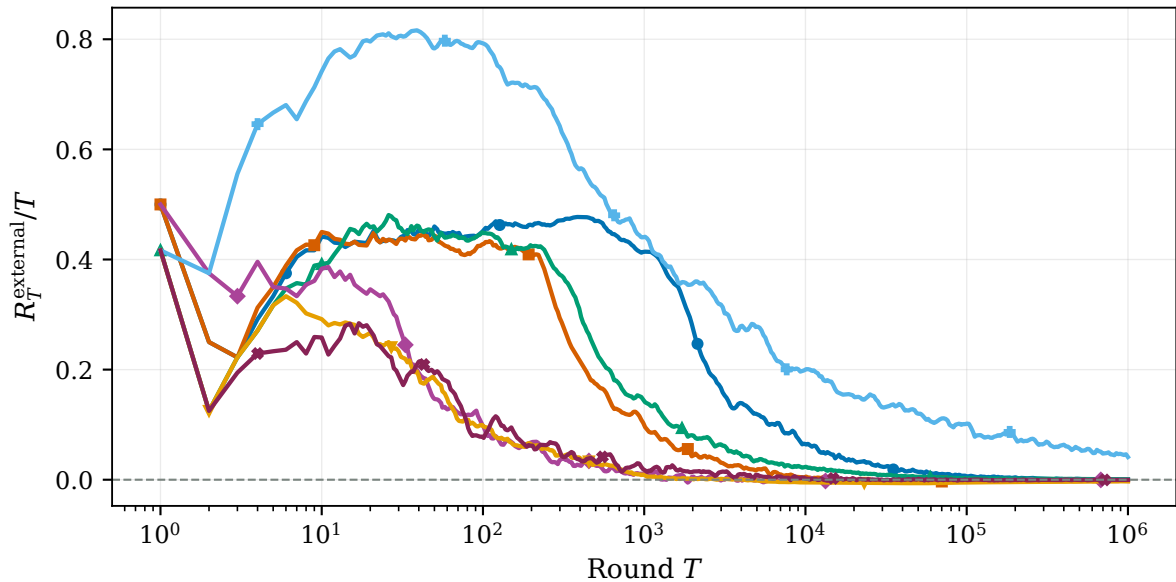


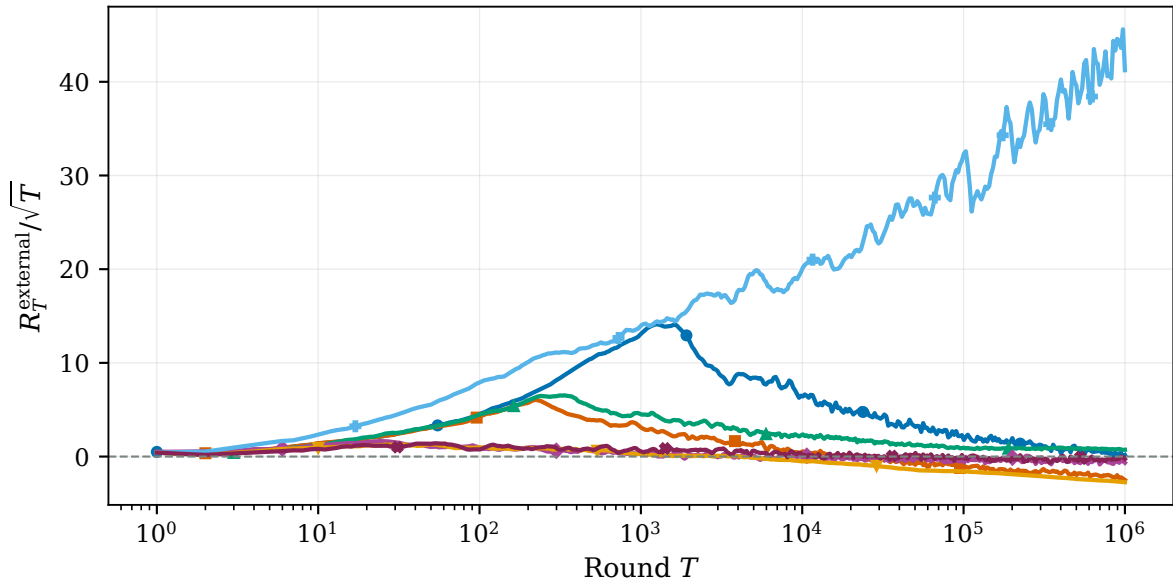
Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithms: BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: External regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge: mean = 0.000175083, SD = 0.000465202, SE = 0.000134292
BM-OptHedge: mean = -0.00249917, SD = 0.00047666, SE = 0.0001376
Hedge: mean = 0.0007245, SD = 0.000326643, SE = 9.42938e-05
Ito-Hedge: mean = -0.000669917, SD = 0.00050081, SE = 0.000144571
OptHedge: mean = -0.00274242, SD = 0.00049422, SE = 0.000142669
RM: mean = 0.0412372, SD = 0.0104722, SE = 0.00302308
SRM: mean = -0.000268167, SD = 0.000794326, SE = 0.000229302



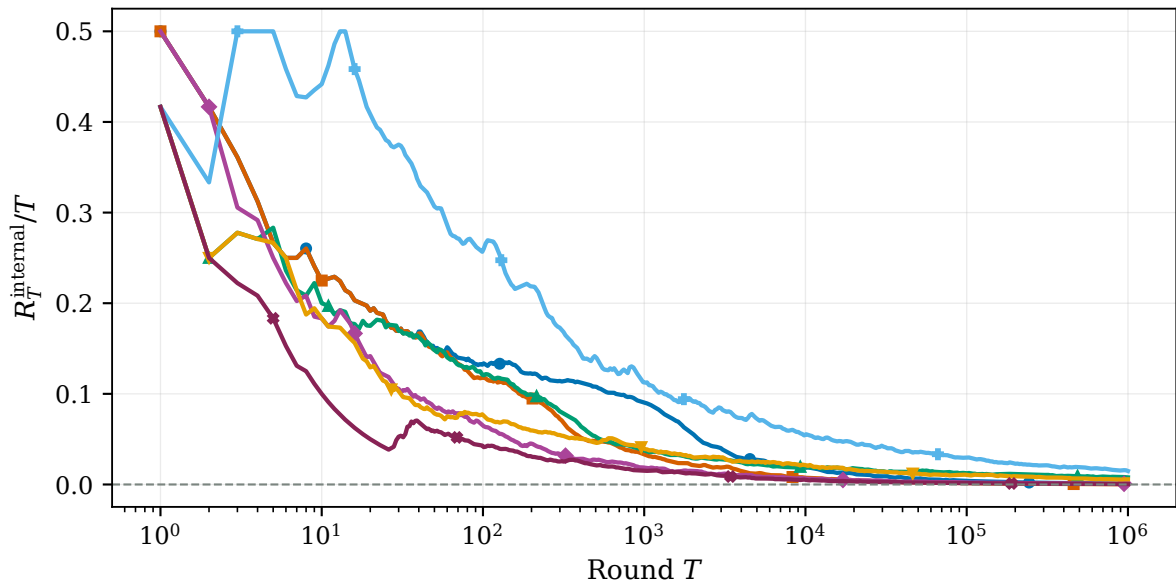
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.175083, SD = 0.465202, SE = 0.134292
BM-OptHedge:	mean = -2.49917, SD = 0.47666, SE = 0.1376
Hedge:	mean = 0.7245, SD = 0.326643, SE = 0.0942938
Ito-Hedge:	mean = -0.669917, SD = 0.50081, SE = 0.144571
OptHedge:	mean = -2.74242, SD = 0.49422, SE = 0.142669
RM:	mean = 41.2372, SD = 10.4722, SE = 3.02308
SRM:	mean = -0.268167, SD = 0.794326, SE = 0.229302



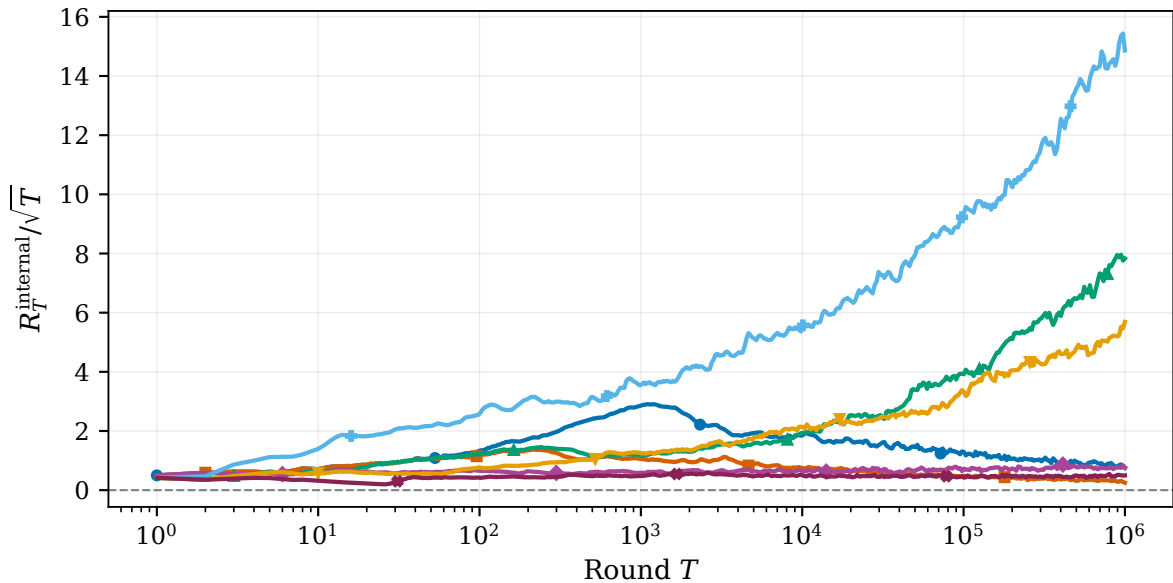
Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithms: BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Internal regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge: mean = 0.000749333, SD = 0.000111192, SE = 3.20984e-05
BM-OptHedge: mean = 0.000249833, SD = 8.82711e-05, SE = 2.54817e-05
Hedge: mean = 0.00783042, SD = 0.00149371, SE = 0.000431196
Ito-Hedge: mean = 0.000771583, SD = 0.000209516, SE = 6.0482e-05
OptHedge: mean = 0.00568767, SD = 0.000769671, SE = 0.000222185
RM: mean = 0.014864, SD = 0.00232601, SE = 0.000671462
SRM: mean = 0.000497833, SD = 8.72695e-05, SE = 2.51925e-05



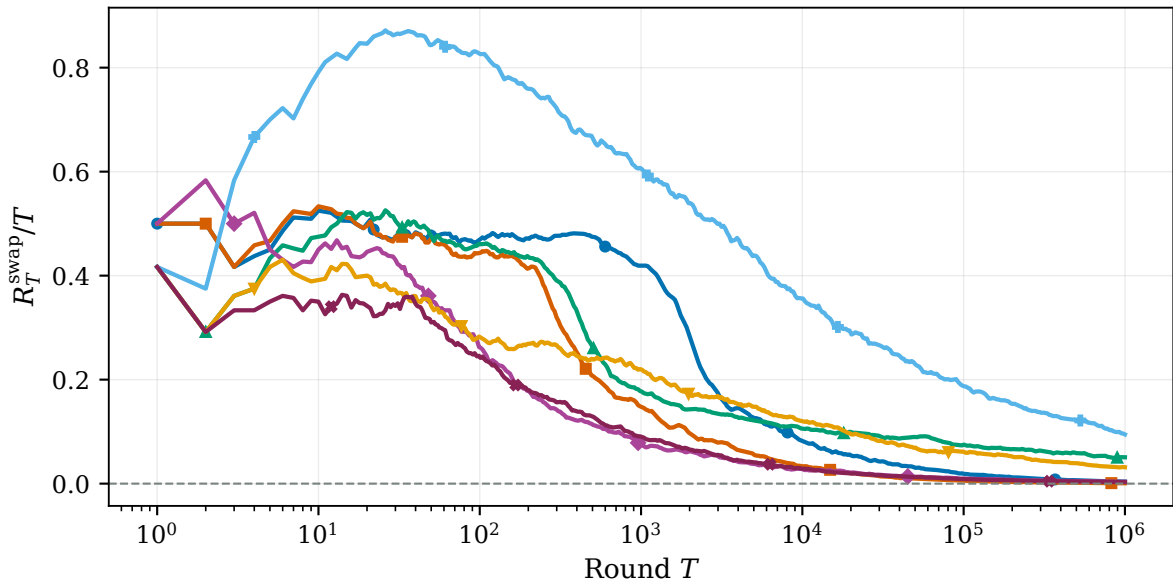
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.749333, SD = 0.111192, SE = 0.0320984
BM-OptHedge:	mean = 0.249833, SD = 0.0882711, SE = 0.0254817
Hedge:	mean = 7.83042, SD = 1.49371, SE = 0.431196
Ito-Hedge:	mean = 0.771583, SD = 0.209516, SE = 0.060482
OptHedge:	mean = 5.68767, SD = 0.769671, SE = 0.222185
RM:	mean = 14.864, SD = 2.32601, SE = 0.671462
SRM:	mean = 0.497833, SD = 0.0872695, SE = 0.0251925



Experiment Information

Figure: Algorithm comparison
Environment: Historical-frequency
Actions: 10
Feedback: Full information
Algorithms: BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon: 1,000,000
Replicates: 12
Seed: 42
Metric: Swap regret
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge: mean = 0.00389417, SD = 0.000468079, SE = 0.000135123
BM-OptHedge: mean = 0.000729333, SD = 0.000297851, SE = 8.59821e-05
Hedge: mean = 0.0506502, SD = 0.00808444, SE = 0.00233378
Ito-Hedge: mean = 0.00349483, SD = 0.000596753, SE = 0.000172268
OptHedge: mean = 0.0315117, SD = 0.00529378, SE = 0.00152818
RM: mean = 0.0945782, SD = 0.00788139, SE = 0.00227516
SRM: mean = 0.00301133, SD = 0.000492072, SE = 0.000142049



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, BM-OptHedge, Hedge, Ito-Hedge, OptHedge, RM, SRM
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 3.89417, SD = 0.468079, SE = 0.135123
BM-OptHedge:	mean = 0.729333, SD = 0.297851, SE = 0.0859821
Hedge:	mean = 50.6502, SD = 8.08444, SE = 2.33378
Ito-Hedge:	mean = 3.49483, SD = 0.596753, SE = 0.172268
OptHedge:	mean = 31.5117, SD = 5.29378, SE = 1.52818
RM:	mean = 94.5783, SD = 7.88139, SE = 2.27516
SRM:	mean = 3.01133, SD = 0.492072, SE = 0.142049

